6.1.7.4 Lateral Pressure on the Superstructure

The lateral pressure due to lateral movement of the earth tends to turn the super structure.

Method of Prevention

The base of the foundation wall should be much wider.

6.1.7.5 Horizontal Movement of the Earth

When buildings are constructed in the low level area and on the river bed where the soil is loose, the foundation may fail due to the horizontal movement of earth.

Method of Prevention

To avoid sliding of soil, bearing walls or pillar with plates may be constructed.

6.1.7.6 Transpiration of Trees and Shrubs

The moisture content of soil is absorbed by the penetration of roots from the trees and plants around the foundation of the building. So cracks are formed due to the shrinkage of soil.

Method of Prevention

i. The foundations should be beyond the roots of tree. Minimum depth should be one meter.

ii. The fast growing trees and trees requiring more water should be 8 meter away from the building.

6.1.7.7 Atmospheric Action

The important factors affecting the foundation are sun, wind and rain. Chemical reactions takes place when the chemical substances in the rain water come in contact with earth, resulting in adverse effects.

Method of Prevention

i. Foundation should be deep upto where the rain water cannot reach.

ii. After the masonry works are finished, sides of wall should be filled with earth and consolidated well. Rain water should be drained properly and it should not stagnate near the walls.



Leaning tower of Pisa, Italy

The tower's tilt began during construction in the 12th century, caused by an inadequate foundation on ground too soft on one side to properly support the structure's weight. The tilt increased in the decades before the structure was completed in the 14th century. It gradually increased until the structure was stabilized by efforts in the late 20th and early 21st centuries



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